A virtual perfusion bioreactor and digital shadow of CAR T cell expansion

Technical documentation of the `closed-loop-cell-therapy` repository

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Abstract

To manufacture chimeric antigen receptor (CAR) T cells of the desired quantity and quality, the cells must be expanded ex vivo for an optimal duration, which in turn requires knowledge of the cell concentration during the expansion period. Egan et al. (2025) developed a digital shadow of CAR T cell expansion that provides a soft sensor of cell concentration in real time: a mechanistic model of cell growth within a proportional-integral-derivative (PID) controlled perfusion bioreactor, fitted to online data of the perfusion rate and the glucose and lactate concentrations. The present work re-implements that digital shadow and, because the original datasets are proprietary, embeds it in a virtual bioreactor that generates online and off-line data for six bioreactor runs of the Aglaris FACER from the published model parameters. The virtual bioreactor adds sensor noise, bubble artefacts, the published low-pass filter, PID feedback on the filtered glucose signal and a mechanical fault. Digital shadow training recovers the yield parameters to within 5%. Real-time testing shows that with 2.5 days of online data the final cell concentration is predicted with a mean relative error of 8% (maximum 23%), compared with 13% reported by Egan et al. As in the original study, the confidence bands are too narrow, because the uncertainty in the fixed parameters Y_{xs} and t_{lag} is not propagated.

Keywords: cell growth, digital shadow, mathematical model, PID control, predictive model, process analytical technology, soft sensor

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1 | INTRODUCTION

Prolonged ex vivo expansion has been associated with progressive differentiation of CAR T cells that can result in less effective treatment, whereas activated CAR T cells must be administered at a sufficiently high dose to be effective [1]. Both the quality and the quantity of CAR T cells are therefore critical quality attributes (CQAs), and the optimal harvest time is patient specific. Hard sensors such as dielectric spectroscopy are less suited to bioreactors in which the cells grow whilst sedimented on a membrane, and resuspending the cells for off-line analysis risks breaching the sterility barrier and can disturb continued cell growth.

Egan et al. [1] addressed this with a digital shadow, defined as a virtual representation with a one-way transfer of data from the physical to the virtual component (in contrast to a digital twin, which transfers data in both directions and can adaptively control the physical system). The digital shadow provides nowcasts and short-term forecasts of the cell concentration from the hard sensors already used to monitor and control the process.

This document describes a re-implementation of that work with two purposes: (i) to reproduce the digital shadow and its training and testing procedures, and (ii) to provide a virtual bioreactor in which the digital shadow, and later extensions such as ensemble-based estimation and supervisory decision support, can be developed and tested without access to the proprietary datasets.

2 | MATERIALS AND METHODS

2.1 | Mathematical

Batch-perfusion process. Cells with concentration $[X(t)]$ are seeded at time t_I and retained in the bioreactor via sedimentation. Perfusion does not begin immediately at t_I but at t_{sed}, after which fresh media with glucose concentration $[S_I]$ is perfused into the bioreactor at a rate $D(t)$, and spent media with glucose and lactate concentrations $[S(t)]$ and $[P(t)]$ is perfused out at the same rate, maintaining a constant volume V. The perfusion rate is

$$D(t) = Q(t)/V, \tag{1}$$

where $Q(t)$ is the volumetric flow rate; $D(t) = 1/\text{d}$ corresponds to one vessel volume per day (VVD). The contents of the bioreactor are modelled as homogeneous; the model is of the unstructured and unsegregated type and is formulated via nonlinear ordinary differential equations (ODEs). Following Egan et al., the following assumptions are made: the flow rate is sufficiently small that only a negligible number of cells are removed in the effluent; glucose and lactate concentrations have a negligible inhibitory effect on cell growth; the dominant inhibitory effect is due to the cells competing with each other to grow in a space-limited environment; cell death and maintenance are negligible; and the growth yield factors are constant.

Cell growth model. The cell concentration and specific growth rate are given by a lag period $t_{\text{lag}} - t_I$ followed by a special case of the Richards growth model [2], applicable in the early stages of cell growth when $[X(t)] \ll [K]$:

$$[X(t)] = \frac{[X_I]}{\left(1 - H(t - t_{\text{lag}})\,\mu_{\max}\,\nu\,(t - t_{\text{lag}})\right)^{1/\nu}} \quad \text{for } t \geq t_I \text{ and } \nu < 0, \tag{2}$$

$$\mu(t) = \frac{H(t - t_{\text{lag}})\,\mu_{\max}}{1 - H(t - t_{\text{lag}})\,\mu_{\max}\,\nu\,(t - t_{\text{lag}})} \quad \text{for } t \geq t_I \text{ and } \nu < 0, \tag{3}$$

where $[X_I]$ is the initial cell concentration, $\mu_{\max}$ is the maximum specific growth rate at t_{lag}, ν determines the decrease of the specific growth rate over time, and H denotes the Heaviside step function (eqs 10, 11 and S60, S61 of [1]). Substituting $\nu = -1$ gives approximately linear cell growth and $\nu = -1/2$ approximately quadratic growth; increasingly small values of ν recover the limiting case of exponential growth (Figure 1).

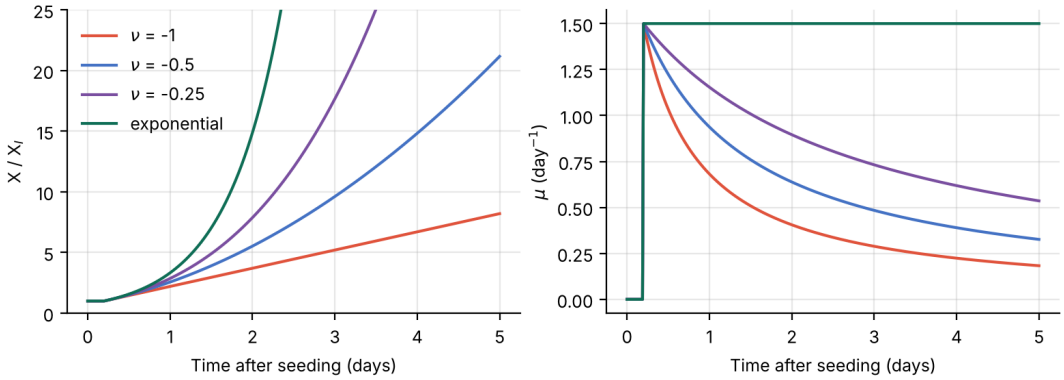


Figure 1: Cell concentration (left) and specific growth rate (right) given by equations (2) and (3) for $\mu_{\max} = 1.5/\mathrm{d}$, $t_{\mathrm{lag}} - t_I = 0.2$ days and four values of ν.

If the final cell concentration $[X_F]$ at time t_F is known, $\mu_{\max}$ can be given as a combination of the other parameters, and equation (2) can be rearranged to give the time t_T at which the cell concentration reaches a threshold concentration $[X_T]$:

$$\mu_{\max} = \frac{1 - ([X_I]/[X_F])^{\nu}}{\nu\,(t_F - t_{\mathrm{lag}})}, \qquad t_T = t_{\mathrm{lag}} + \frac{1 - ([X_I]/[X_T])^{\nu}}{\mu_{\max}\,\nu} \quad \text{for } \nu < 0. \tag{4}$$

Glucose, lactate and PI controlled perfusion rate. The glucose and lactate concentrations and the perfusion rate are modelled as

$$\frac{\mathrm{d}[S]}{\mathrm{d}t} = H(t - t_{\mathrm{sed}})\,D(t)\,([S_I] - [S(t)]) - \frac{\mu(t)[X(t)]}{Y_{xs}} \quad \text{for } t \geq t_I, \tag{5}$$

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = Y_{px}\,\mu(t)[X(t)] - H(t - t_{\mathrm{sed}})\,D(t)\,[P(t)] \quad \text{for } t \geq t_I, \tag{6}$$

$$\frac{\mathrm{d}D}{\mathrm{d}t} = H(t - t_{\mathrm{sed}})\left(K_p\,\frac{\mathrm{d}([S_{\mathrm{sp}}] - [S(t)])}{\mathrm{d}t} + K_i\,([S_{\mathrm{sp}}] - [S(t)])\right) \quad \text{for } t \geq t_I, \tag{7}$$

where Y_{xs} and Y_{px} denote the yield of cells from glucose and the yield of lactate from cells, respectively, $[S_{\mathrm{sp}}]$ is the glucose concentration set-point, and K_p and K_i are the proportional and integral gains (eqs 2–4 of [1]). Equation (7) models the PID controller as a proportional-integral (PI) controller, a good approximation since the derivative gain K_d is 3–4 orders of magnitude smaller than K_p and K_i. The yield of lactate from glucose is

$$Y_{ps} = Y_{px}\,Y_{xs}. \tag{8}$$

Assuming that, for $t \geq t_{sp}$, the glucose concentration remains approximately at its set-point, $\mathrm{d}[S]/\mathrm{d}t \approx 0$, and the lactate concentration tends to the steady state

$$[P_{ss}] = Y_{ps}\,([S_I] - [S_{\mathrm{sp}}]), \tag{9}$$

which is the yield of lactate from glucose multiplied by the difference between the glucose concentration in the fresh media and the glucose concentration set-point (eq. 18 of [1]).

Dissolved oxygen. For the extended model, the normalised dissolved oxygen (DO) $C'(t) = [C(t)]/[C^*]$ is modelled as

$$\frac{\mathrm{d}C'}{\mathrm{d}t} = \left(H(t - t_{\mathrm{sed}})\,D(t) + k_L a\right)\left(1 - C'(t)\right) - \frac{\mu(t)[X(t)]}{Y'_{xc}} \quad \text{for } t \geq t_I, \tag{10}$$

where $k_L a$ is the volumetric mass transfer coefficient and $Y'_{xc} = Y_{xc}[C^*]$ is the scaled yield of cells from DO (eqs 8, 9 of [1]).

Discrete solution. The perfusion rate, glucose concentration and lactate concentration have a lower bound of zero. Following Egan et al., equations (5)–(7) and (10) are approximated by forward finite difference equations and solved via the forward Euler method with a step size of 10 s:

$$[S(t_{k+1})] = \max\Big(0,\ [S(t_k)] + \Delta t\Big(D(t_k)([S_I] - [S(t_k)]) - \frac{\mu(t_k)[X(t_k)]}{Y_{xs}}\Big)\Big), \tag{11}$$

$$D(t_{k+1}) = \max\Big(D_{\min},\ D(t_k) - K_p\big([S(t_{k+1})] - [S(t_k)]\big) + \Delta t\, K_i\big([S_{\mathrm{sp}}] - [S(t_k)]\big)\Big), \tag{12}$$

$$[P(t_{k+1})] = \max\Big(0,\ [P(t_k)] + \Delta t\big(Y_{px}\,\mu(t_k)[X(t_k)] - D(t_k)[P(t_k)]\big)\Big), \tag{13}$$

with $t_0 = t_{\mathrm{sed}}$ and $D(t_0) = D_{\min}$ (eqs S22–S24). $[X(t)]$ and $\mu(t)$ are evaluated analytically from equations (2) and (3). The implementation is `process/facer.py`.

2.2 | Virtual bioreactor

Culture system. The virtual bioreactor represents the large chamber of the Aglaris FACER, a completely closed and automated cell production system with a single-use cartridge of two culture chambers (volumes of 13.4 mL and 134 mL, surface areas of 10 cm^2 and 100 cm^2). Within the culture chambers, cells are cultivated under static conditions on a gas-permeable membrane that separates the controlled media zone from the controlled gas zone.

Process dynamics. The virtual bioreactor (`sim/runner.py`) evolves equations (2), (3), (5), (6) and (10) with the step size of 10 s and the parameter values of a given bioreactor run. Unlike the digital shadow, the perfusion rate is set by a PID controller (`control/perfusion.py`) acting on the filtered online glucose signal, in the velocity form

$$\begin{gathered} D(t_{k+1}) = \max\Big(D_{\min},\ D(t_k) + K_p(e_{k+1} - e_k) + K_i\,\Delta t\, e_k + K_d\,\frac{e_{k+1} - 2e_k + e_{k-1}}{\Delta t}\Big), \\ e_k = [S_{\mathrm{sp}}] - [\tilde{S}(t_k)], \end{gathered} \tag{14}$$

where $[\tilde{S}]$ denotes the filtered glucose measurement; with $K_d = 0$ and $[\tilde{S}] = [S]$ this reduces to equation (12).

Online data. Glucose and lactate concentrations are measured every 10 s via enzymatic sensors, with additive Gaussian noise (standard deviation 0.08 mmol/L). Bubbles generated during the expansion process interfere with the online sensors, leading to artificial drops in the glucose and lactate concentration readings followed by rises once the bubbles have moved away. These are modelled as a Poisson process (3 per day) in which the readings fall to a fraction of 0.3–0.8 of the true concentration for 20–300 s. To prevent these fluctuations impacting the perfusion rate, the raw glucose and lactate data are filtered before being passed to the PID controller by a lowpass Butterworth filter of order 2 with a cut-off frequency of 0.001 Hz, applied causally to the 10 s readings. Not all bubble artefacts are filtered out, which results in downward spikes of the glucose (and lactate) concentration followed by upward spikes in the perfusion rate (Figure 3). Normalised DO is recorded with additive noise (standard deviation 0.01). The noise levels and bubble statistics are assumed values; the filter specification is that of Egan et al.

Off-line data. Accurately measuring the cell concentration requires the cells to be in suspension, and resuspending the cells after they have sedimented can disturb their growth. Off-line cell counts are therefore generated only at t_I and t_F, with a coefficient of variation of 5%.

Mechanical issue. In run 3, a mechanical issue occurred 2.5 days following the start of perfusion and lasted for approximately 0.5 days, during which perfusion stopped and no online data of the glucose and lactate concentrations were recorded. This is reproduced by setting $D(t) = 0$ and withholding all online data over $[t_{\mathrm{sed}} + 2.5,\ t_{\mathrm{sed}} + 3.0]$ days.

Bioreactor runs. Table 1 lists the observed (input) and best-fit (output) parameters of the six runs, taken from Tables 1, 2 and S4 of [1] (`process/parameters.py`). All runs share $D_{\min} = 0.067/\mathrm{d}$,

$K_p = 0.215\,\mathrm{L/(mmol\,day)}$, $K_i = 5.16\,\mathrm{L/(mmol\,day^2)}$, $K_d = 7.46 \times 10^{-5}\,\mathrm{L/mmol}$ and $V = 134\,\mathrm{mL}$. In runs 1–5, cells were expanded in TexMACS medium with IL-7 and IL-15 (media A); in run 6, in Advanced TCM with IL-2 (media B). Run 4 expanded patient CAR T cells; the others, donor T cells. Run 3 was the independent test dataset of the original study and has no best-fit parameters; it is assigned the mean training values, with $\mu_{\max}$ given by equation (4) such that $[X(t_F)] = [X_F]$.

Table 1: Observed and best-fit parameters of the six bioreactor runs in the large chamber of the Aglaris FACER [1]. Times in days from the start of the process; concentrations in mmol/L and 10^6 cells/mL; Y_{xs} in cells/pmol; Y_{px} in pmol/cell; k_La in 1/d; Y'_{xc} in 10^6 cells/mL. Italics: assigned values (run 3).

Run	t_I	t_{sed}	t_F	$[S_I]$	$[S_{\mathrm{sp}}]$	$[X_I]$	$[X_F]$	Y_{xs}	Y_{px}	$t_{\mathrm{lag}} - t_I$	ν	$\mu_{\max}$	k_La	Y'_{xc}
1	0.11	0.34	5.05	25	20	1.49	19.0	0.231	7.93	0.160	−0.922	2.14	3.00	3.68
2	0.14	1.02	6.03	25	20	0.37	19.0	0.275	6.25	0.172	−0.477	2.02	741	0.0220
3	0.11	0.86	6.01	25	20	1.49	32.5	*0.259*	*7.62*	*0.092*	*−0.537*	*1.36*	*3.00*	*3.68*
4	2.09	2.84	7.00	25	20	0.62	10.9	0.272	9.09	0.128	−0.524	1.39	1.73	6.27
5	1.92	2.92	6.95	25	17	1.13	8.73	0.207	6.83	0.000	−0.000	0.405	42.9	0.189
6	0.08	0.83	4.85	26	21	1.49	18.2	0.311	7.99	0.000	−0.761	1.58	20.2	0.893

2.3 | Computational

Digital shadow. The digital shadow (`shadow/fit.py`) is fitted simultaneously to the perfusion rate, glucose concentration and lactate concentration data of each run; the three online data sources are given equal weight and no scaling is applied. Equations (2), (3) and (11)–(13) are used in the fitting process. Fitting is performed by nonlinear least-squares minimisation with the trust-region reflective method of SciPy, which accommodates the parameter bounds $Y_{xs}, Y_{px} > 0$, $0 \le t_{\mathrm{lag}} - t_I < t_F - t_I$ and $\nu < 0$ (Egan et al. used the Levenberg–Marquardt method in the lmfit package). To reduce computation, the model is evaluated on a 60 s grid and compared with the online data at the same time points.

Digital shadow training. For training purposes, data for both the initial and final cell concentrations constrain the system via equation (4), so that $\mu_{\max}$ does not have to be fitted directly; the fitted parameters are Y_{xs}, Y_{px}, t_{lag} and ν.

Digital shadow testing. In a real-time scenario, either the lag period or the maximum specific growth rate must be known a priori for the model to be structurally identifiable (Table S1 of [1]); fixing the lag period alone leaves the model practically unidentifiable. The lag period and the yield of cells from glucose are therefore fixed at their mean training values ($t_{\mathrm{lag}} - t_I = 0.092$ days, $Y_{xs} = 0.259$ cells/pmol), and Y_{px}, $\mu_{\max}$ and ν are fitted to the online data available at the time of model fitting. The real-time version is fitted with 1.5 to 4 days of online data (in half-day time steps) from the start of perfusion; for run 3, with 1.5 to 2.5 days, prior to the mechanical issue. The extended model including equation (10) is not used for fitting, since k_La and Y'_{xc} are not practically identifiable from these data (Table S4 of [1]).

Confidence bands. Confidence bands for $[X(t)]$ and $\mu(t)$ are derived from the partial derivatives with respect to $\mu_{\max}$ and ν (eqs S67–S70) and the covariance matrix resulting from the fitting process:

$$\Delta f = \nabla f^{\mathsf{T}}\,\mathrm{cov}[\mu_{\max}, \nu]\,\nabla f, \qquad B = \hat{f} \pm \tau_{\alpha/2,n}\sqrt{\chi^2_n\,\Delta f}, \tag{15}$$

where $f = \{[X(t)], \mu(t)\}$, n is the number of degrees of freedom, χ^2_n is the reduced chi-squared statistic and $\tau_{\alpha/2,n}$ is the upper $\alpha/2$ critical value of Student's t distribution, with $\alpha = 0.05$ (eqs S71, S72).

3 | RESULTS

3.1 | Virtual bioreactor data

Figure 2 shows the simulated online and off-line data of run 1 in the layout of Figure 3 of Egan et al. After the sedimentation period, the glucose concentration falls below its set-point, the PI(D) controller increases the perfusion rate, and following an initial transient the glucose concentration remains at 20 mmol/L with an approximately constant perfusion rate, consistent with the approximately linear cell growth for $\nu \approx -1$. The lactate concentration approximately stabilises at a steady state shortly after the glucose concentration stabilises at its set-point, in agreement with equation (9). For all six runs, the simulated final cell concentration equals the published $[X_F]$, the values of $\mu_{\max}$ and of the required expansion period $t_T - t_I$ for $[X_T] = 10^7$ cells/mL agree with Table 2 of [1] to within 0.015 day^{-1} and 0.04 days, respectively, and the terminal lactate concentration is within 5% of $[P_{ss}]$.

Figure 4 shows run 3. During the mechanical issue, perfusion stops, glucose is depleted and lactate accumulates; on resumption, the controller drives the perfusion rate above the trajectory of the undisturbed model before the glucose concentration returns to its set-point. Cell growth is unaffected, in accordance with the model assumption of negligible substrate and product inhibition.

3.2 | Digital shadow training

Table 2 shows the best-fit parameters for the five training runs. The yields are recovered to within 5% (Y_{xs}) and 7% (Y_{px}) of the parameters used to generate the data, and ν and the lag period are recovered closely. The deviations are attributable to the 5% off-line counting error, which propagates into Y_{xs} through $\mu_{\max}$, and to bubble artefacts, which bias the filtered lactate signal downwards and hence Y_{px}. The reduced chi-squared values are below unity because the residuals are dominated by the filtered (and therefore smoothed) sensor noise.

Table 2: Best-fit parameters with 95% confidence intervals for the five virtual training runs, and the values used to generate the data in square brackets (Table 2 of [1]).

Run	Y_{xs}	Y_{px}	$t_{\text{lag}} - t_I$	ν	χ^2_n
1	0.241 (0.236, 0.246) [0.231]	7.51 (7.36, 7.66) [7.93]	0.162 (0.047, 0.278) [0.160]	−0.895 (−0.905, −0.886) [−0.922]	0.11
2	0.282 (0.281, 0.282) [0.275]	6.02 (6.00, 6.03) [6.25]	0.171 (0.145, 0.197) [0.172]	−0.473 (−0.478, −0.468) [−0.477]	0.11
4	0.282 (0.280, 0.283) [0.272]	8.65 (8.61, 8.69) [9.09]	0.124 (0.079, 0.169) [0.128]	−0.527 (−0.536, −0.518) [−0.524]	0.08
5	0.215 (0.214, 0.216) [0.207]	6.52 (6.49, 6.56) [6.83]	0.000 (0.000, 0.072) [0.000]	−0.011 (−0.026, 0.000) [−0.000]	0.06
6	0.321 (0.318, 0.324) [0.311]	7.63 (7.55, 7.70) [7.99]	0.000 (0.000, 0.072) [0.000]	−0.762 (−0.772, −0.752) [−0.761]	0.09

3.3 | Digital shadow testing

Figure 5 shows the relative errors of the predicted final cell concentration with varying quantities of online data, and the cell concentration predictions based on 2.5 days of online data (cf. Figures 5 and 6 of [1]); Table 3 lists the values. With 1.5 days of online data, five of the six runs are predicted to within a 30% relative error. With 2.5 days (or more) of online data, the final cell concentration is predicted to within a 30% relative error for all runs, with a mean relative error of 8% (Egan et al.: 13%, standard deviation $\approx$ 6%).

The confidence bands are too narrow, in that the final cell concentration is frequently not within the band. Runs 1 and 6 are consistently over- and under-predicted, respectively, irrespective of the quantity of online data. Their yields of cells from glucose (0.231 and 0.311 cells/pmol) differ from the fixed mean value (0.259 cells/pmol) by -11% and $+20\%$; since the digital shadow infers the growth of cells from their glucose consumption via Y_{xs}, this deviation propagates directly into the predicted cell concentration and is not captured by the confidence bands of equation (15).

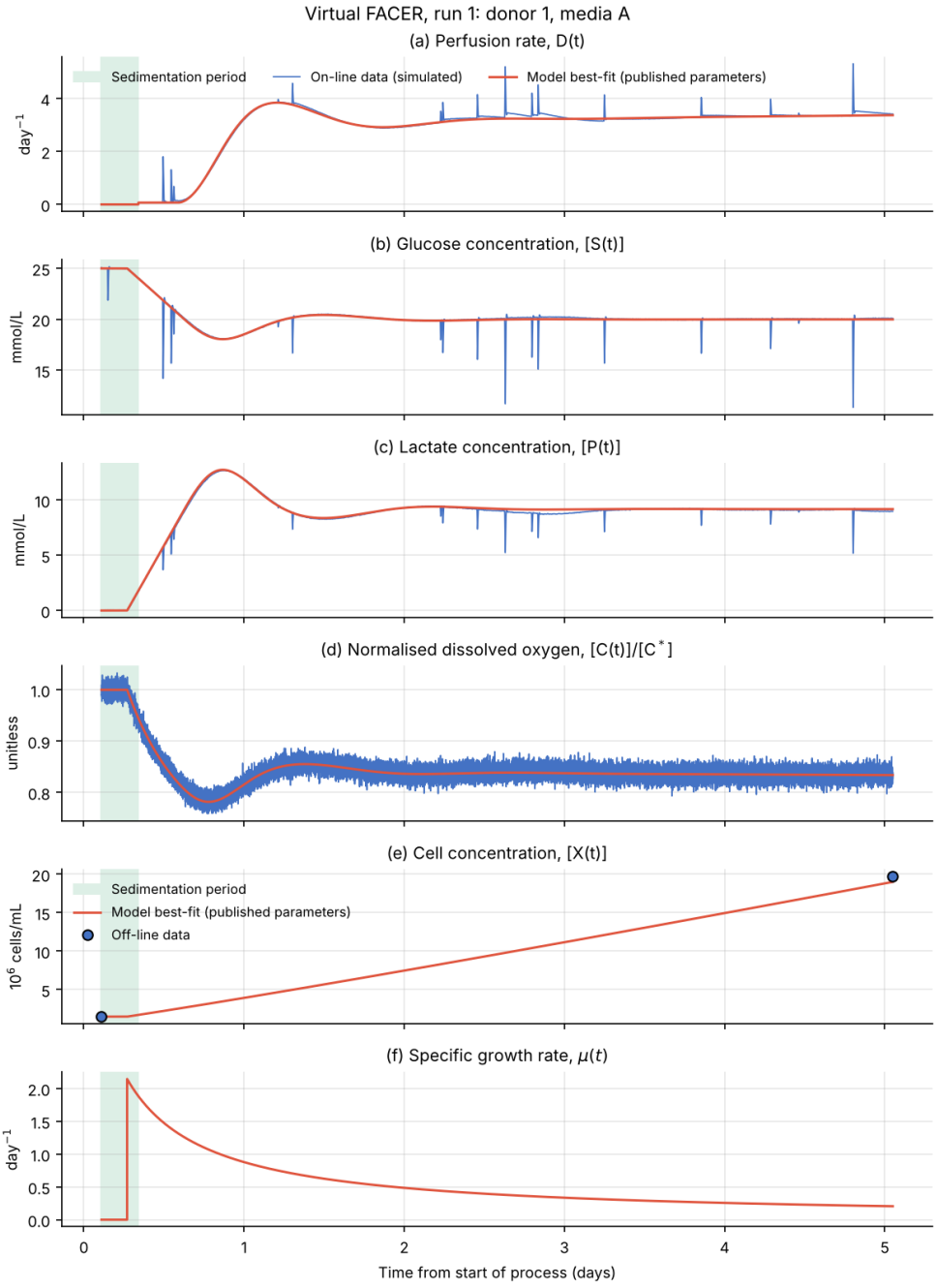


Figure 2: Key data from virtual run 1 in the large chamber of the FACER and model trajectories with the published best-fit parameters (cf. Figure 3 of [1]).

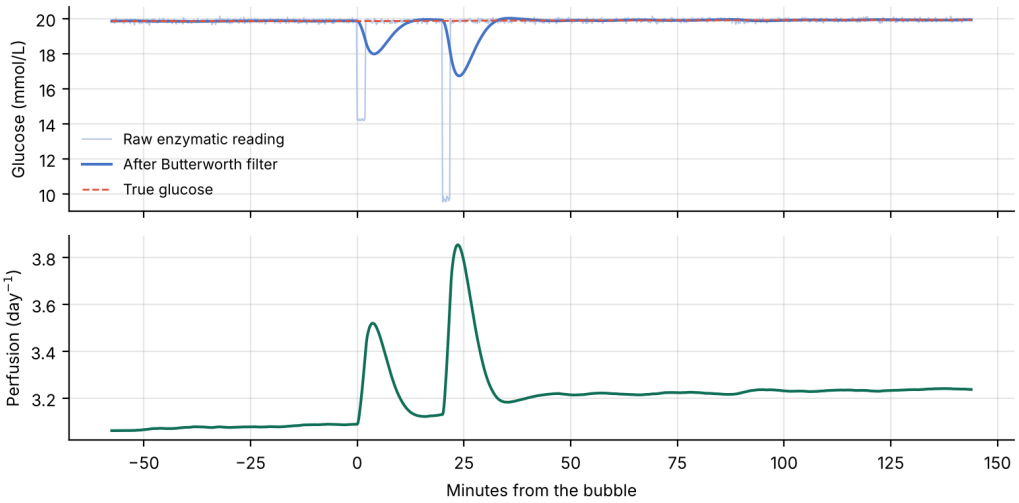


Figure 3: Two bubble artefacts in virtual run 1: raw and filtered glucose concentration (top) and the resulting upward spikes in the perfusion rate (bottom).

Table 3: Relative error (%) of the predicted final cell concentration.

Online data (days)	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Mean absolute
1.5	28.2	12.7	−4.6	0.6	−5.3	−31.9	13.9
2.0	18.6	2.0	−4.0	−0.9	−3.9	−26.3	9.3
2.5	15.8	−1.5	−3.9	−5.5	−0.6	−22.7	8.3
3.0	12.0	−3.7	–	−5.9	4.4	−20.2	9.2
3.5	9.2	−6.9	–	−6.4	10.5	−19.2	10.4
4.0	7.7	−7.6	–	−7.4	15.7	−17.8	11.3

4 | DISCUSSION

The virtual bioreactor reproduces the published parameter values, the qualitative features of the published online data (the transient response of the PI(D) controller, the stabilisation of the glucose concentration at its set-point, the steady state of the lactate concentration, and the bubble artefacts that survive the filter) and the mechanical issue of run 3. Within this environment, the re-implemented digital shadow reproduces the main findings of Egan et al.: the training parameters are recovered, and with at least 2.5 days of online data the final cell concentration is predicted with a relative error of less than 30%.

As in the original study, the confidence bands should be interpreted as minimum confidence bands, because other sources of uncertainty are not captured: the uncertainty in Y_{xs} and t_{lag}, the uncertainty in the online and off-line measurements, and the uncertainty in the model structure. Egan et al. proposed fitting a bivariate distribution to the parameter estimates of Y_{xs} and t_{lag} from many runs, repeatedly sampling from that distribution and using the sampled values when fitting to the online data in real time, to obtain a distribution of predicted cell concentrations. This is an ensemble approach, and an ensemble Kalman filter formulation of the digital shadow, in which Y_{xs} and t_{lag} are treated as uncertain augmented states, is a natural extension.

The strongest assumption of the model is that the media and cells are homogeneously mixed, although the cells are sedimented on the membrane and the influent and effluent are situated higher up. The model further assumes that all cells are homogeneous; T cells comprise $CD4^+$ and $CD8^+$ subsets with different expansion profiles, which an extension to a segregated model could address. In the virtual bioreactor, these assumptions hold by construction, so the reported errors reflect parameter and measurement uncertainty only, and not model structure uncertainty.

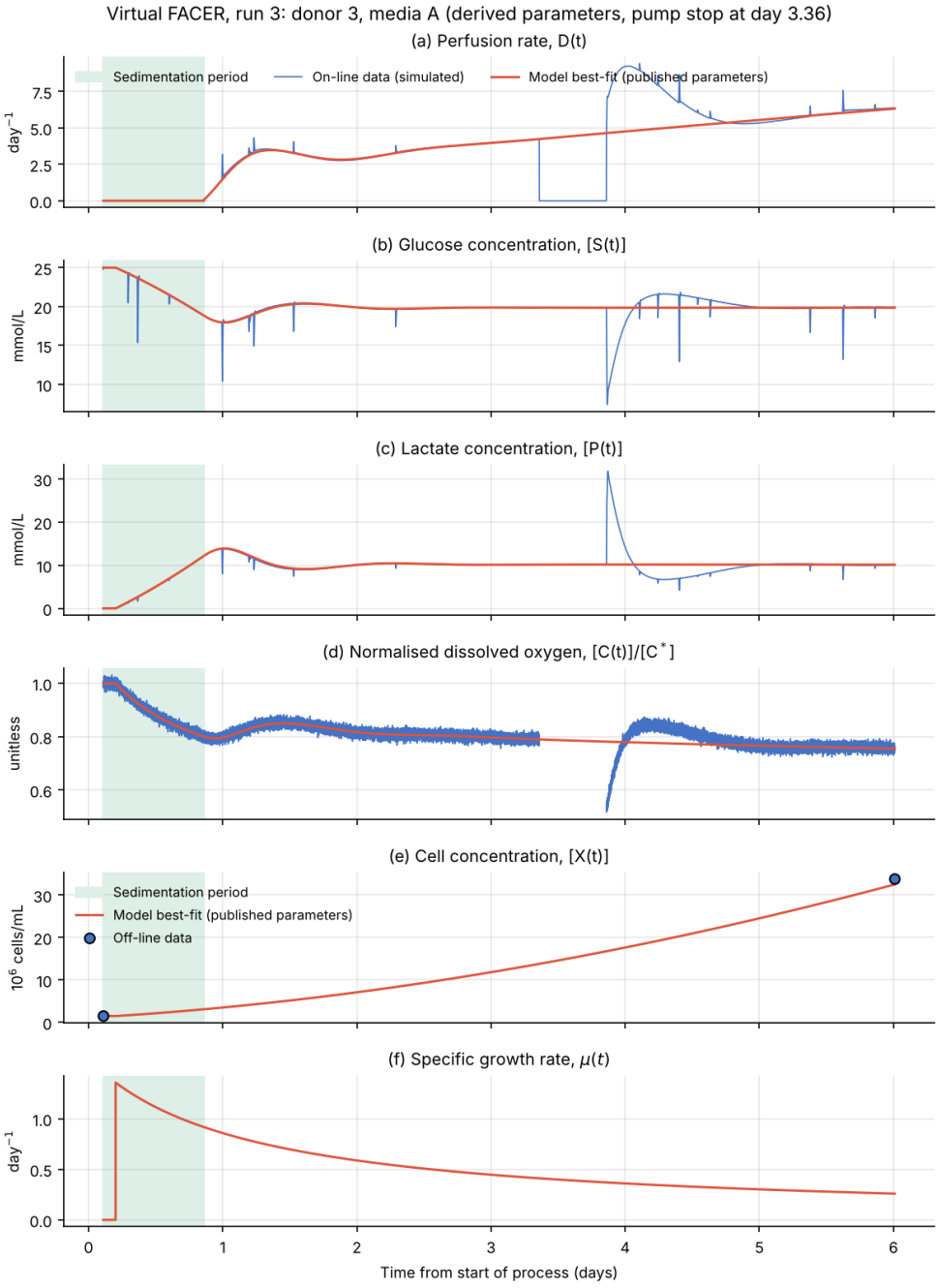


Figure 4: Key data from virtual run 3 with the mechanical issue 2.5 days following the start of perfusion, during which no online data were recorded.

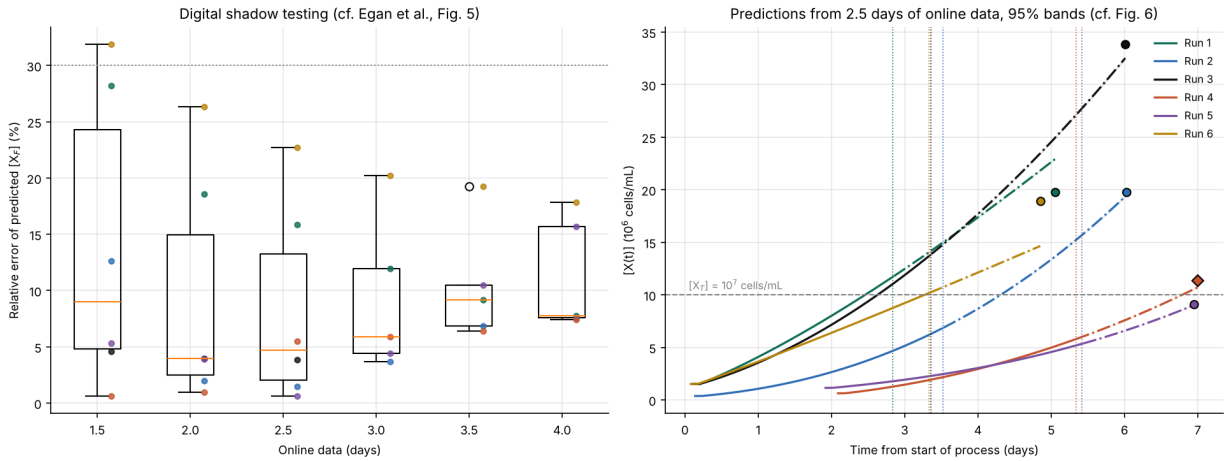


Figure 5: Left: relative errors of predicted final cell concentration with varying quantities of online data. Right: cell concentration predictions based on 2.5 days of online data and the off-line initial cell concentration; solid lines are predictions up to the time of fitting, dash-dotted lines future predictions, dotted vertical lines the time of fitting and symbols the off-line final cell concentrations (diamond: patient 1).

Other limitations are specific to this re-implementation: the online data are synthetic, since the original datasets are proprietary; the noise levels and bubble statistics are assumed values; and the parameters of run 3 are assigned rather than estimated.

5 | CONCLUSION

A digital shadow of CAR T cell expansion in a PID controlled perfusion bioreactor has been re-implemented, together with a virtual bioreactor that generates online and off-line data from the published parameters of six bioreactor runs of the Aglaris FACER. The digital shadow provides nowcasts and short-term forecasts of the cell concentration, and hence of the time at which a threshold cell concentration is reached, using only the hard sensors already used to monitor and control the process. The virtual bioreactor provides a test environment for further work: exposing the bioreactor via industrial communication protocols, an ensemble-based digital shadow that captures the uncertainty in Y_{xs} and t_{lag}, decision support for the harvest time with the human operator in the loop, and tuning of the PID parameters via the digital shadow, which Egan et al. identify as a means to improve the stability of the glucose concentration around its set-point.

NOMENCLATURE

Symbol	Description	Units
t	time from the start of the process	days
t_I, t_{sed}, t_F	initial (seeding), final sedimentation and final perfusion times	days
t_{lag}	time at which cell growth starts; $t_{\text{lag}} - t_I$ is the lag period	days
t_T	time at which $[X(t)]$ reaches $[X_T]$	days
$[X(t)]$, $[X_I]$, $[X_F]$, $[X_T]$	cell concentration; initial, final and threshold values	10^6 cells/mL
$[S(t)]$, $[S_I]$, $[S_{\text{sp}}]$	glucose concentration; in fresh media; set-point	mmol/L
$[P(t)]$, $[P_{ss}]$	lactate concentration; steady state	mmol/L
$C'(t) = [C(t)]/[C^*]$	normalised dissolved oxygen	unitless
$D(t)$, D_{min}	perfusion rate; minimum perfusion rate	day^{-1}

$Q(t)$, V	volumetric flow rate; bioreactor volume	mL/day; mL
$\mu(t)$, $\mu_{\max}$	specific growth rate; maximum specific growth rate	day^{-1}
ν	decrease of specific growth rate	unitless
$[K]$	carrying capacity	10^6 cells/mL
Y_{xs}	yield of cells from glucose	cells/pmol
Y_{px}	yield of lactate from cells	pmol/cell
Y_{ps}	yield of lactate from glucose	unitless
$k_L a$, Y'_{xc}	volumetric mass transfer coefficient; scaled yield of cells from DO	day^{-1}; 10^6 cells/mL
K_p, K_i, K_d	proportional, integral and derivative gains	L/(mmol day); L/(mmol day^2); L/mmol
H	Heaviside step function	–
χ^2_n	reduced chi-squared	unitless

APPENDIX: IMPLEMENTATION AND VERIFICATION

`process/growth.py`	equations (2)–(4), (9); sensitivities S67–S70
`process/parameters.py`	Table 1; mean training values
`process/facer.py`	equations (11)–(13) and (10); the model fitted by the digital shadow
`control/perfusion.py`	PID controller, equation (14)
`control/pid.py`	generic PI controller with anti-windup (control refresher notes)
`sensors/online.py`	enzymatic sensors, bubble artefacts, Butterworth filter, DO, off-line counts
`sim/runner.py`	virtual bioreactor, mechanical issue, data export
`shadow/fit.py`	digital shadow training and testing, confidence bands, t_T
`scripts/simulate_run.py`, `scripts/realtime_test.py`	Figures 2, 4 and 5; Table 3

The test suite (`pytest`, 31 tests) verifies the special cases of the growth model, $\mu(t) = \mathrm{d}\ln[X(t)]/\mathrm{d}t$, the agreement of $\mu_{\max}$ and $t_T - t_I$ with Table 2, the mutual consistency of equation (4), the analytical sensitivities against finite differences, the final cell concentrations, set-point tracking and lactate steady state of every run, the absence of perfusion during the sedimentation period, the filter and bubble models, the mechanical issue of run 3, the recovery of the training parameters, the 30% bound of digital shadow testing with 2.5 days of online data, and the narrowing of the confidence bands with increasing online data.

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